

Probability Is Not an Answer

A model holds two memories. Which one comes out is not a question of size.



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The problem

How a language model handles a fact is studied by **five separate communities** that rarely cite one another.

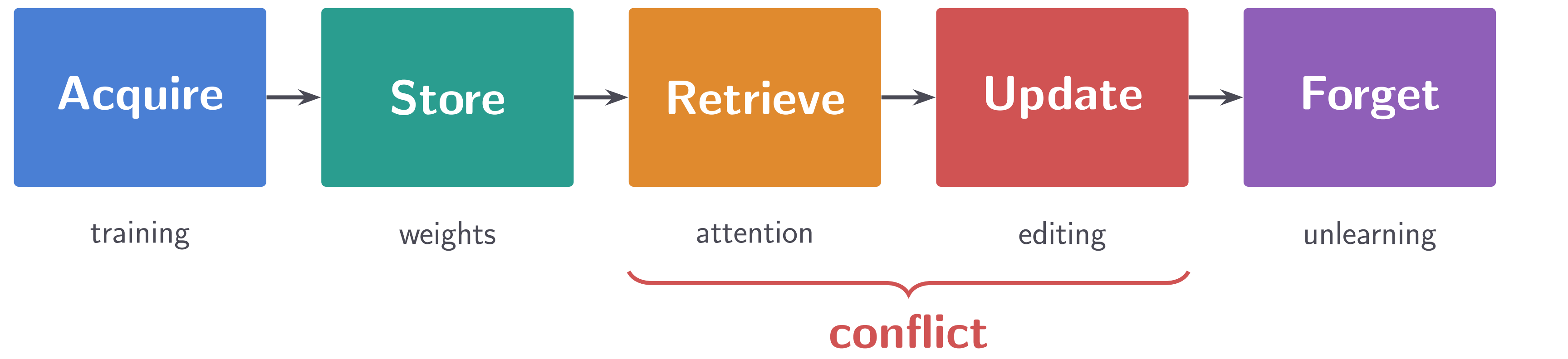
- retrieval-augmented generation
- knowledge editing
- continual learning
- context engineering
- agent memory

Each has its own benchmarks, and each passes them. A system can pass **all five** and still fail, because nothing tests what happens **between** them.

Across **20 representative papers**, the median work covers **two** of the five stages. The boundaries are where deployment breaks.

The framework

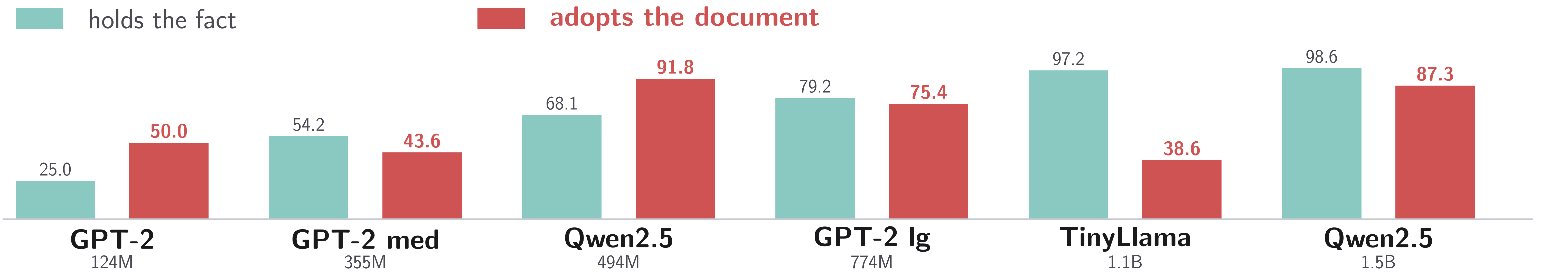
A fact passes through five stages. Naming them as one lifecycle is what makes the boundaries between them visible.



The failure below sits between **Retrieve** and **Update**: the document arrives, and nothing tells the model which copy of the fact to trust. That boundary belongs to neither field beside it, which is why nobody tests it.

The finding

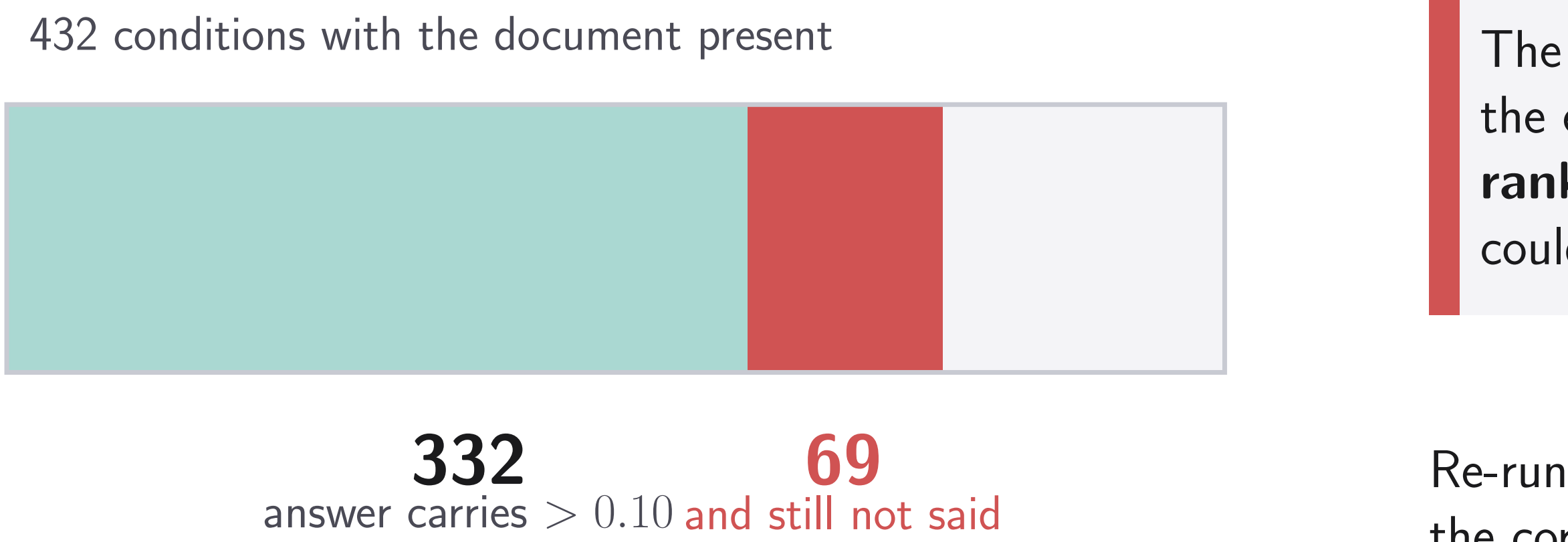
Give each model a fact it already holds, then a document asserting something else. **864 forward passes**: six models, 24 counterfactual probes, three phrasings, with and without the document. Read at the **rank** of the answer, which is what the model would actually say.



Holding a fact scales with size. Using the document does not.
Knowing the answer climbs monotonically with parameter count, 25.0 to 98.6 per cent, with no inversion anywhere. What the model then does with a contradicting document follows no such order: TinyLlama holds the fact 97.2 per cent of the time and defers least of the six, while Qwen2.5-0.5B, a third its size, defers 91.8 per cent.

Probability is not an answer

The probability a model puts on an answer and the answer it gives are **different quantities**. Mass is spread over tens of thousands of tokens, so the answer has only to be ranked first, not to be large.



In **20.8%** of the conditions where the in-context answer carries real probability, it is **still not what the model says**. A reader given only the probability would conclude the document had landed.

Recording the rank costs **one comparison**.

A correction

An earlier version of this work reported that no model answers a drug-withdrawal probe correctly under any phrasing. **That claim was false**, and how it failed is why this poster exists.

The script behind it recorded only the **surprisal** of the expected answer. It never recorded that answer's **rank**, nor which token the model put first, so it could not have tested the claim it was cited for.

Re-run with rank recorded, **three of the six** models rank the corrected answer first under two of the three phrasings.

A second defect was structural. Those probes were **real-world updates**, so whether a model ever held the stale fact depended on its training cutoff, and the six cutoffs differ. The withdrawal predates every model's training data, so **no conflict existed to resist**.

Reproduce it

Every number on this poster comes from **one deterministic script**. Nothing is sampled, so the digits come out identical on any machine.

```
pip install torch transformers
python counterfactual_update.py
```

It runs the six models in float32 on **CPU**, so the digits do not depend on a particular accelerator's arithmetic. About twenty minutes on a laptop.

Every figure on this poster is **generated** from the measurement, never typed. The earlier version of this work quoted a number its own code had never produced, which is the failure that guards against.

Six models, 24 facts, three phrasings. The probes are all national capitals, so the result is about one kind of fact; whether numeric or temporal facts behave the same way is untested.